REP-037 · Proposal: data insights for balance, retention and online health

Game	Fruit Market
Report date	2026-10-27 17:02 (Asia/Kuala_Lumpur)
Type	Plan proposal (data handling change). **Nothing built yet**
Request	"can we track this please, for balance changes, user retention etc, generally data insights"
Proposed plan	PLAN-018 (awaiting approval)

1. What we have today

- Anonymous daily totals, no IDs, opt-out in Settings, kept 90 days (`docs/ANALYTICS.md`):
  - app opens, installs per week and return days (retention);
  - games started and finished, days reached and quit days;
  - wins and losses, and final cash buckets;
  - tutorial steps, online room creation and errors.
- **Gaps:**
  - nothing about balance (traits, specialities, computer levels, prices, spoilage);
  - counts don't say which rules version a game used;
  - nobody reads them regularly;
  - this session can't read them (no key, server not reachable from here).

2. Proposal

a. New anonymous counters (same privacy rules: totals only, no IDs, opt-out respected)

Question it answers	Counters (each tagged with the rules version)
Which set-ups do people play?	length, players, computer levels, options (Core, Street prices, Market Days, pulse)
Is a trait too strong?	your trait, and whether you won with it
Are specialities worth building?	specialities built (and on which day), and wins with each
Are computer levels right?	wins and losses against Easy, Normal and Hard
Is the price duel working?	per game: share of sales near the customer's top price (bucket); fruit spoiled (bucket); final winning margin (bucket)
Which favour cards matter?	favour cards played
Is online healthy?	matches finished vs abandoned; computer takeovers; quick match found vs gave up, and wait time (bucket)

Question it answers	Counters (each tagged with the rules version)
Did a release change retention?	app version on each app start

b. Weekly insights report

- A GitHub Action reads the live and preview stats with your `STATS_KEY` (stored as a GitHub secret), once a week and on demand.
- It builds a one-page report:
  - return rates on day 1, 7 and 30;
  - first-game and tutorial funnel, and where people quit;
  - win rates by trait, speciality and computer level, each with its sample size;
  - balance compared with the computer-player simulations;
  - errors.
- Each balance change is then judged on real players by rules version, not only on simulations.

c. Paperwork

- `docs/ANALYTICS.md` and `docs/privacy.html` list every new counter.
- Tests prove nothing identifying can be sent and that opt-out sends nothing.

3. Cost and risk

- **Money:** no new service or dependency. More counter writes on Cloudflare, still small (batched). Whether that stays free depends on your Cloudflare plan, which is still unconfirmed.
- **Privacy:** unchanged in kind (anonymous totals). Small groups can still be revealing, so the report hides any number under 5.
- **Limit:** counters can show *what* happens, not *why* or whether it's fun. Watched playtests are still needed.
- **Live game:** the counters only reach live players after the next release (separate publish approval).

4. Questions for you

1. **Where should the weekly report go?** (a) A private GitHub Actions download, recommended; (b) saved into the repo's docs folder, visible to anyone with repo access.
2. **How long to keep the data?** 90 days today. For trends across releases, 365 days is better (still anonymous).
3. **Will you add `STATS_KEY` as a GitHub secret?** I'll give step-by-step instructions; without it, the report can't run automatically.